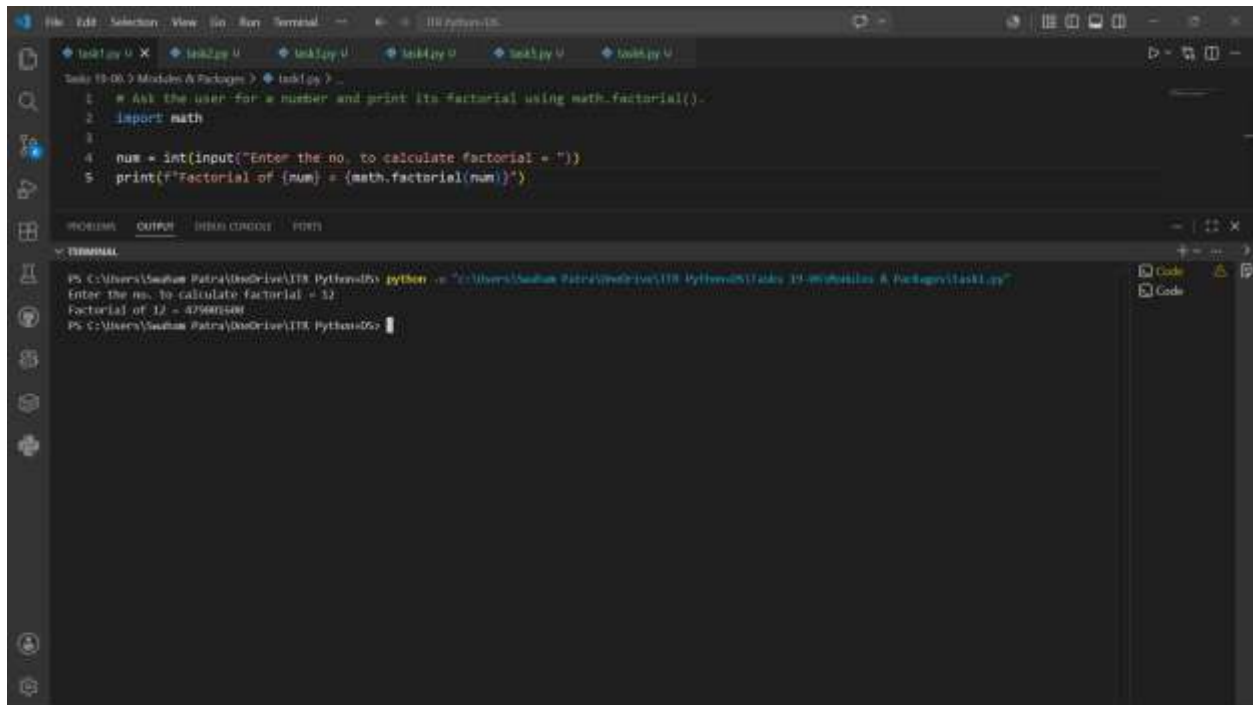


Swaham Pitambar Patra

Modules & Packages :

Task 1:



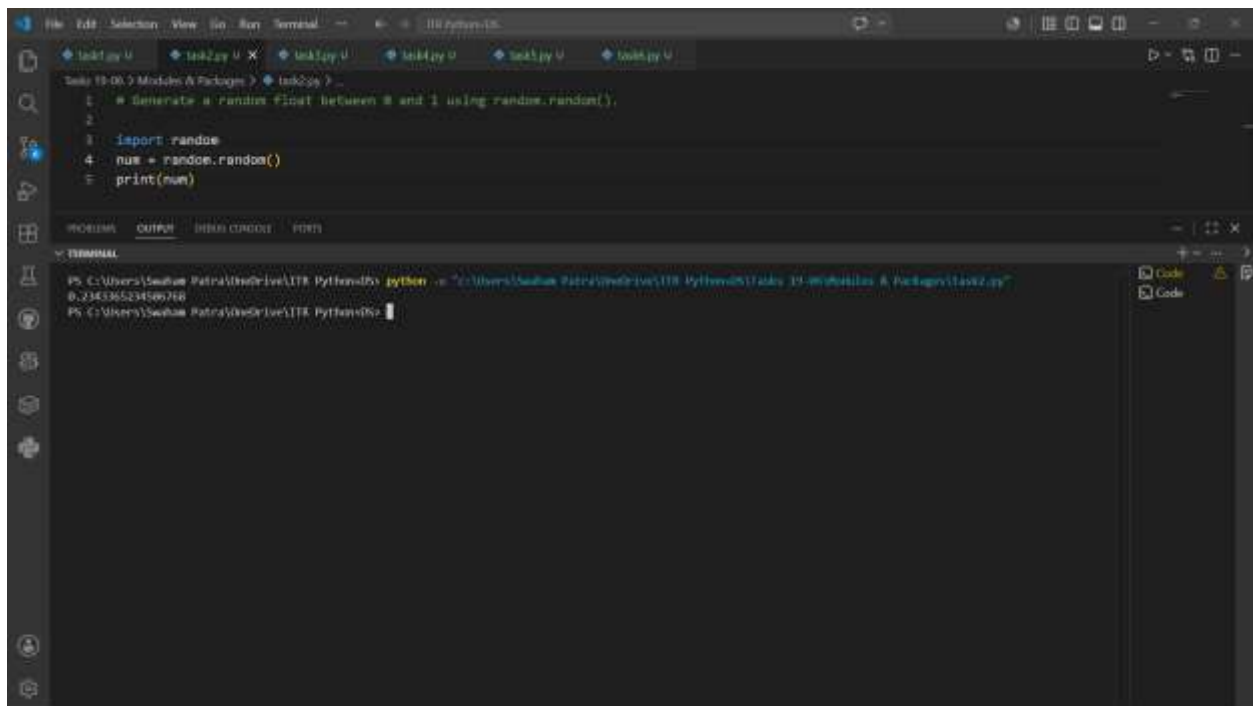
The screenshot shows a Python IDE with a file named 'task1.py'. The code in the editor is as follows:

```
1 # Ask the user for a number and print its factorial using math.factorial().
2 import math
3
4 num = int(input("Enter the no. to calculate factorial = "))
5 print(f"Factorial of {num} = {math.factorial(num)}")
```

The terminal output shows the execution of the script:

```
PS C:\Users\Swaham Patra\OneDrive\IT\ Python> python -i "C:\Users\Swaham Patra\OneDrive\IT\ Python\Task 1\ Modules & Packages\task1.py"
Enter the no. to calculate factorial = 12
Factorial of 12 = 479001600
PS C:\Users\Swaham Patra\OneDrive\IT\ Python>
```

Task 2:



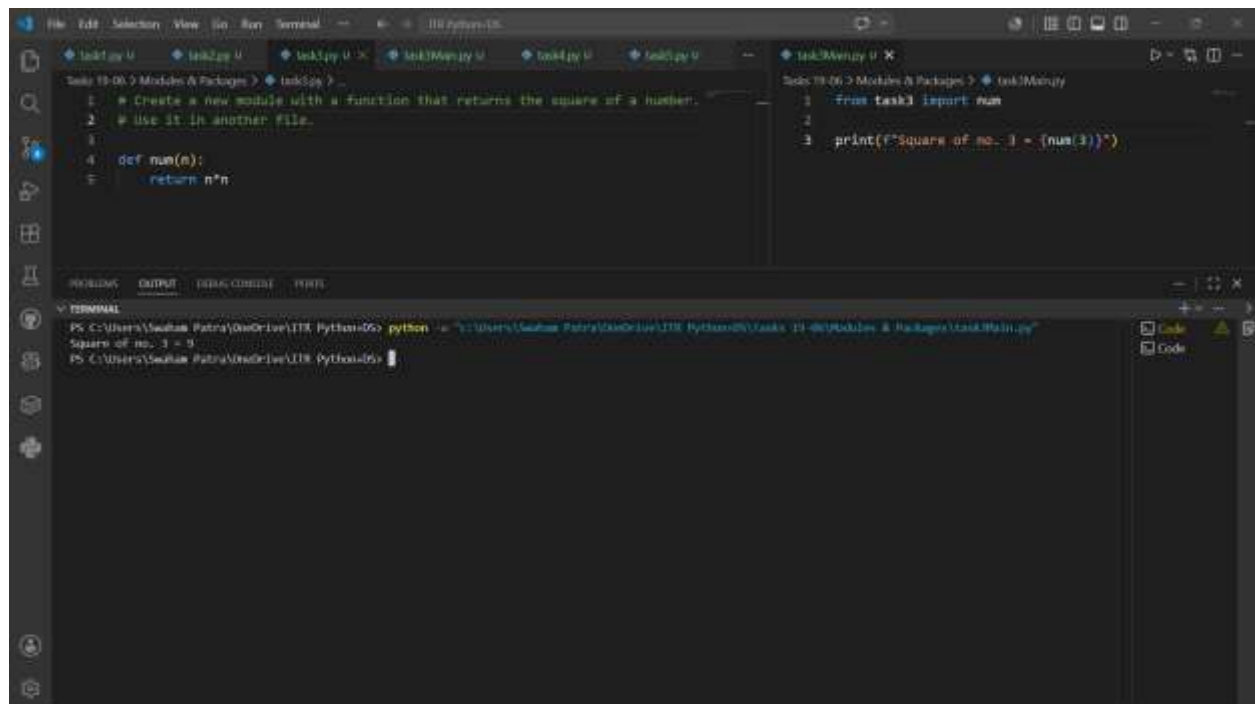
The screenshot shows a Python IDE with a file named 'task2.py'. The code in the editor is as follows:

```
1 # Generate a random float between 0 and 1 using random.random().
2
3 import random
4 num = random.random()
5 print(num)
```

The terminal output shows the execution of the script:

```
PS C:\Users\Swaham Patra\OneDrive\IT\ Python> python -i "C:\Users\Swaham Patra\OneDrive\IT\ Python\Task 2\ Modules & Packages\task2.py"
0.2343365234586768
PS C:\Users\Swaham Patra\OneDrive\IT\ Python>
```

Task 3:



```
task3.py
1 # Create a new module with a function that returns the square of a number.
2 # Use it in another file.
3
4 def num(n):
5     return n*n
```

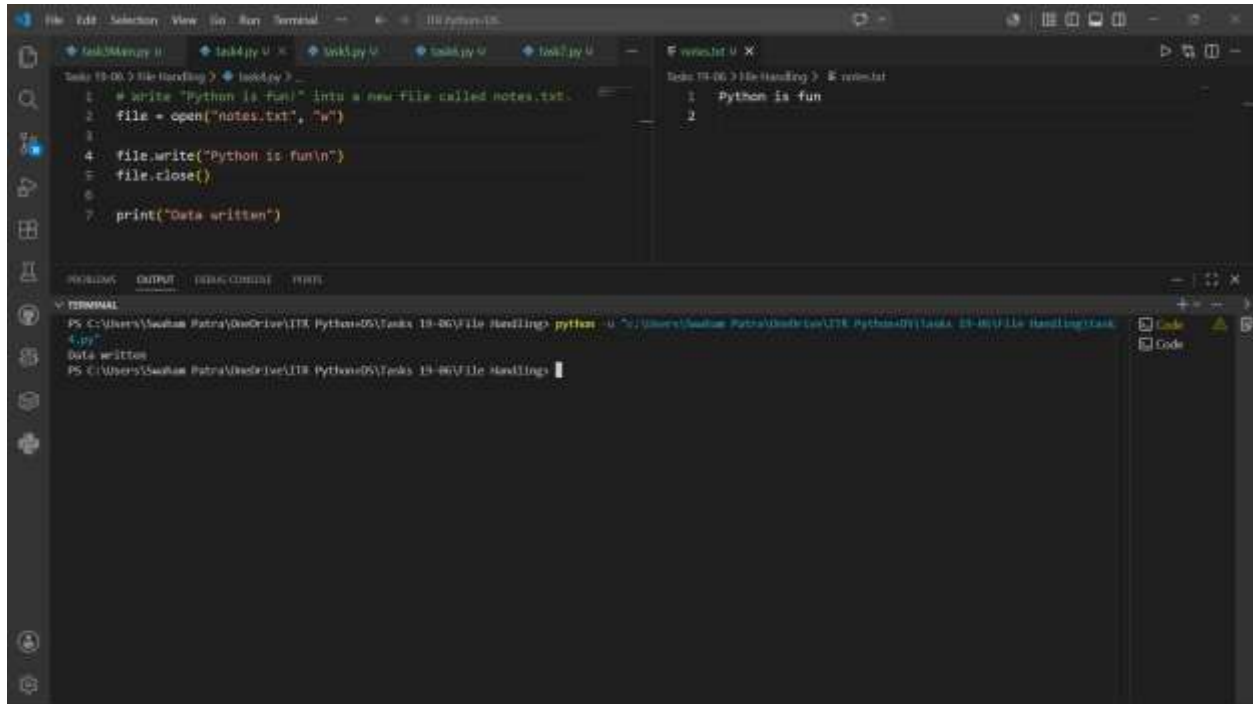
```
task3Main.py
1 from task3 import num
2
3 print(f"Square of no. 3 = {num(3)}")
```

TERMINAL

```
PS C:\Users\Saahaa Patra\Desktop\ITE Python\05> python -i "C:\Users\Saahaa Patra\Desktop\ITE Python\05\task3-19-06\Modules & Packages\task3Main.py"
Square of no. 3 = 9
PS C:\Users\Saahaa Patra\Desktop\ITE Python\05>
```

File Handling :

Task 4:

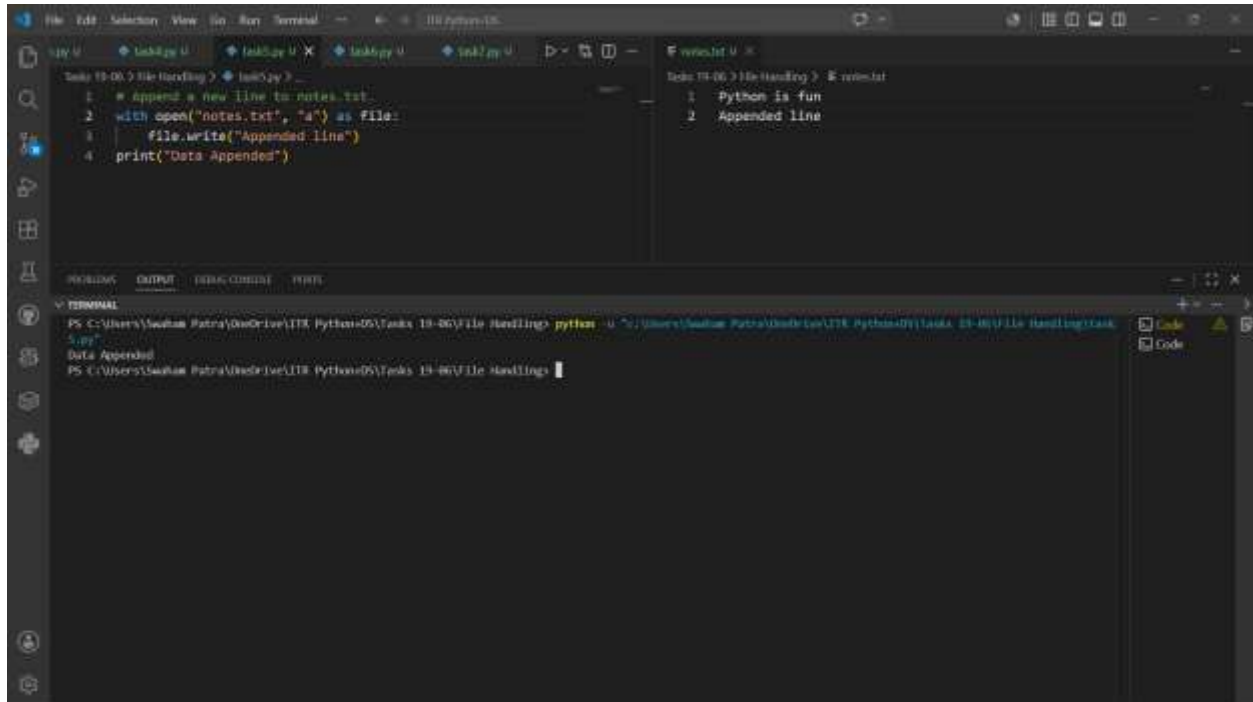


```
Task 19-06 > File Handling > task4.py > ...
1 # Write "Python is fun!" into a new file called notes.txt.
2 file = open("notes.txt", "w")
3
4 file.write("Python is fun\n")
5 file.close()
6
7 print("Data written")
```

```
Task 19-06 > File Handling > notes.txt
1 Python is fun
2
```

```
TERMINAL
PS C:\Users\Swahus Patra\OneDrive\ITK Python\05\Tasks 19-06\File Handling> python 4.py
Data written
PS C:\Users\Swahus Patra\OneDrive\ITK Python\05\Tasks 19-06\File Handling>
```

Task 5:

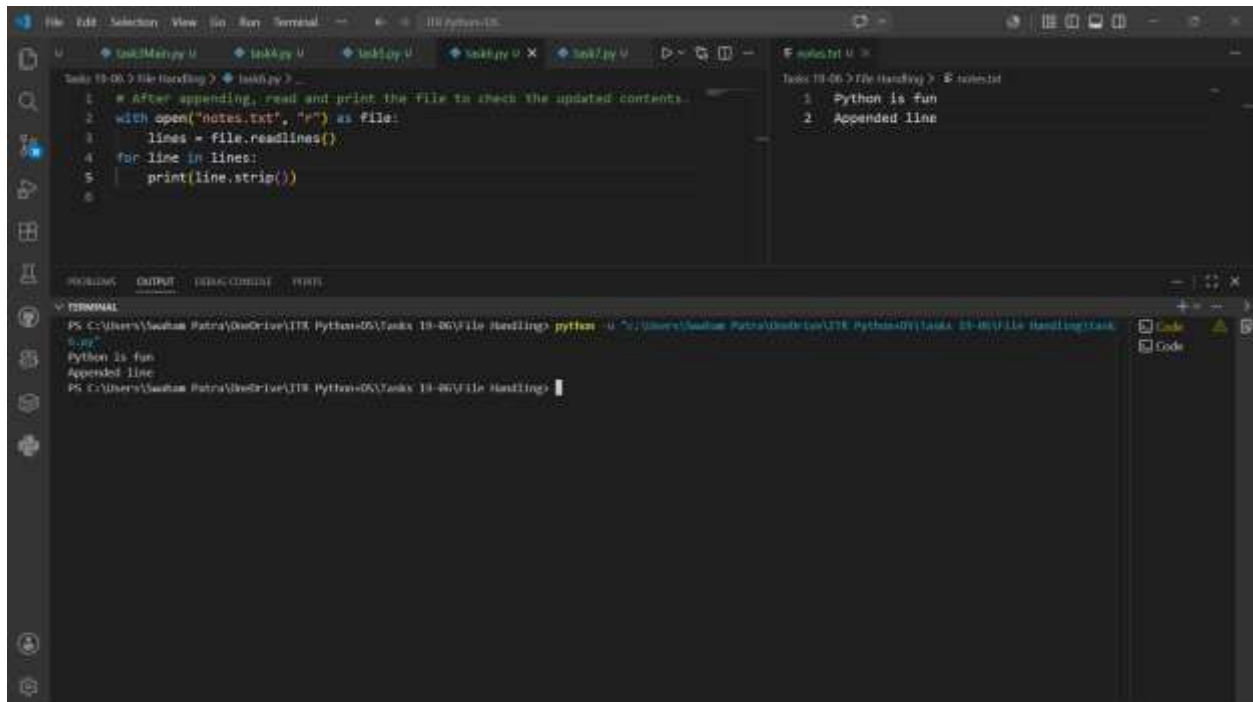


```
Task 19-06 > File Handling > task5.py > ...
1 # Append a new line to notes.txt.
2 with open("notes.txt", "a") as file:
3     file.write("Appended line")
4 print("Data Appended")
```

```
Task 19-06 > File Handling > notes.txt
1 Python is fun
2 Appended line
```

```
TERMINAL
PS C:\Users\Swahus Patra\OneDrive\ITK Python\05\Tasks 19-06\File Handling> python 5.py
Data Appended
PS C:\Users\Swahus Patra\OneDrive\ITK Python\05\Tasks 19-06\File Handling>
```

Task 6:

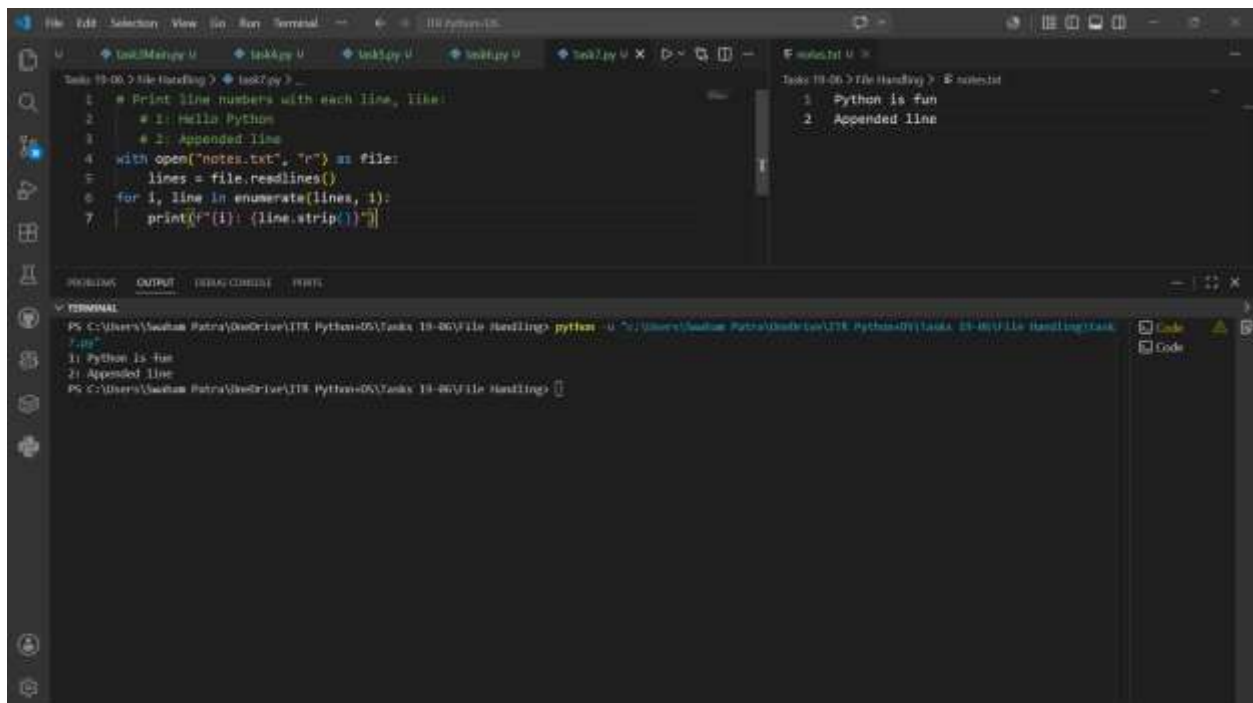


The screenshot shows the Visual Studio Code editor with a Python file named `task6.py` and a terminal window. The code in `task6.py` reads the contents of `notes.txt` and prints each line. The terminal output shows the execution of the script, which prints "Python is fun" and "Appended line".

```
1 # After appending, read and print the file to check the updated contents.
2 with open("notes.txt", "r") as file:
3     lines = file.readlines()
4     for line in lines:
5         print(line.strip())
6
```

```
PS C:\Users\Saahaa Patra\Documents\ITR Python\05\Tasks 19-06\File Handling> python -u "C:\Users\Saahaa Patra\Documents\ITR Python\05\Tasks 19-06\File Handling\task6.py"
Python is fun
Appended line
PS C:\Users\Saahaa Patra\Documents\ITR Python\05\Tasks 19-06\File Handling>
```

Task 7:



The screenshot shows the Visual Studio Code editor with a Python file named `task7.py` and a terminal window. The code in `task7.py` reads the contents of `notes.txt` and prints each line with its line number. The terminal output shows the execution of the script, which prints "1: Python is fun" and "2: Appended line".

```
1 # Print line numbers with each line, like:
2 # 1: Hello Python
3 # 2: Appended line
4 with open("notes.txt", "r") as file:
5     lines = file.readlines()
6     for i, line in enumerate(lines, 1):
7         print(f"{i}: {line.strip()}")

```

```
PS C:\Users\Saahaa Patra\Documents\ITR Python\05\Tasks 19-06\File Handling> python -u "C:\Users\Saahaa Patra\Documents\ITR Python\05\Tasks 19-06\File Handling\task7.py"
1: Python is fun
2: Appended line
PS C:\Users\Saahaa Patra\Documents\ITR Python\05\Tasks 19-06\File Handling>
```